

Code-Layout, Readability and Reusability

Date	7 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4-space indentation followed consistently across app.py and train_model.py
2	Proper File Structure	Files and folders are logically organized	Yes	Project split into dataset/, templates/, static/, notebook/ folders clearly
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Names like applicant_income, loan_amount, GENDER_MAP, best_model are self-explanatory
4	Function / Method Names	Functions are descriptively named	Yes	Functions named home(), predict(), train_evaluate() clearly describe their purpose
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings on every function, section headers like # Load Dataset, # Handle Missing Values used throughout
6	Modular Design	Code is split into reusable functions/modules	Yes	train_evaluate() reused for all 4 models; Flask routes separated as individual functions
7	No Redundant Code	Duplicate or unused code is removed	Yes	Label encoding loop reuses single LabelEncoder for all columns; no repeated logic
8	Error Handling	Exceptions and errors are handled gracefully	Yes	try-except block in predict() catches all errors and returns a friendly error page

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	train_evaluate() function	Python / Scikit-learn	Reused for all 4 ML models (Decision Tree, Random Forest, KNN, XGBoost) in train_model.py	High

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
2	model.pkl (Saved XGBoost Model)	Python / Pickle	Loaded in app.py for prediction without retraining	High
3	Label Encoding Maps (GENDER_MAP, MARRIED_MAP, etc.)	Python / Dictionary	Used in predict() to encode all form inputs consistently	High
4	style.css (Stylesheet)	CSS	Applied to both index.html and result.html templates	Medium
5	Navbar & Footer HTML blocks	HTML / Jinja2 (Flask)	Same navbar and footer structure reused in index.html and result.html	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	The project follows a well-structured folder hierarchy with organized source code, templates, static files, and dataset.
Readability	5	The code is easy to read with meaningful variable names, proper indentation, and clear function definitions.
Reusability	5	Reusable modules, trained model, HTML templates, CSS, and prediction logic are effectively used throughout the project.
Documentation / Comments	5	The code includes appropriate comments and is well documented, making it easy to understand and maintain.
Overall Score	5 / 5	The Smart Lender project demonstrates excellent code quality, readability, reusability, and maintainability by following good programming practices.